M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo

Full recording stitch

Open full recording

Clock orbit: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right · Effective 3× speed (video encoded at 1.5×, player at 2×). Open orbit comparison

Clock task: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right. Task replay uses the latest orbit mesh under the temporary demo policy. Open task comparison

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

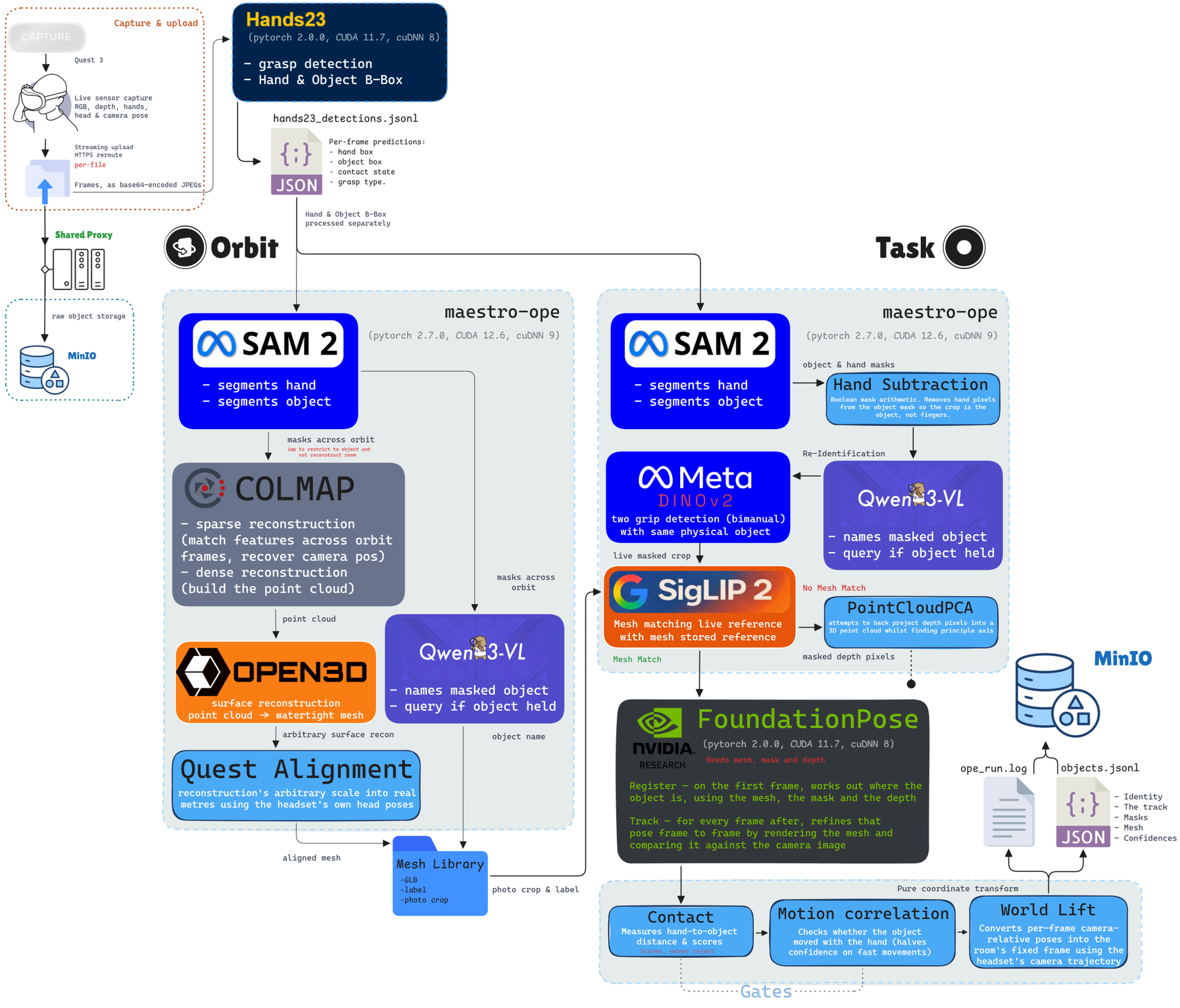


Figure 1. Current Maestro VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

1

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

Quest sensor capture

2

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

MinIO object storage

3

Hands23 hand detection

Provides hand/object contact evidence and candidate boxes before object processing begins.

Hands23 code and explanation

4

SAM2 segmentation

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

SAM2 overview and code

5a

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

Project pose measurements

5b

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

FoundationPose overview and code

6

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

Qwen3-VL-4B-Instruct model

7

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

DINOv2 ViT-S/14 model · SigLIP base model

8

COLMAP mesh reconstruction

Reconstructs object meshes from orbit capture frames for canonical mesh-backed replay.

COLMAP tutorial and code

9

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

Unity XR Interaction Toolkit

System Status

Confirmed steps come first, followed by demo patches, partial fixes and disabled work, then outcomes still to confirm. Open up to two cards at once. The side connections show where a working step still depends on unfinished follow-up. Cards marked “Update, 12 Oct” were corrected after later tests: see the audit below, with videos.

Hands23: the starting point

A detected grip supplies the hand and object boxes that seed the rest of the pipeline.

Module 3 **Confirmed**

> **Update, 12 Oct:** The detector weights were lost in a server reboot, restored and moved to permanent storage, then checked against a reference run (identical results). See the audit below.

The live worker checks the separate Hands23 service and saves its detections before object processing. Missing detections now raise an error, instead of being mistaken for a recording with no object. A brief deliberate grip seeds the orbit route; task recordings use contact evidence to start tracking.

These boxes changed the architecture: SAM2 can now isolate both the hand and the object, so a separate SAM3 hand detector is no longer needed. A box is the starting region, not a finished mask or mesh.

Hands23 code and explanation

SAM3 removal: SAM2 is sufficient for the current prompts

Hands23 supplies the boxes; SAM2 creates the object and hand masks.

Modules 3, 4 **Confirmed**

The earlier reference-image mesh route needed hand pixels removed before generating a mesh. SAM3 was also used for an object-presence check. It was removed after the Hands23-guided SAM2 route was evaluated; this was more than removing a duplicate model.

Model	Plain-language role	Why this pipeline changed
SAM2	Select a region using a point or box, then follow its pixels through video.	Hands23 supplies an object box and a hand box. SAM2 makes separate object and hand masks; hand pixels are subtracted.
SAM3	Find and segment concepts using text or example prompts.	Previously used for hand removal and an object-presence check. Its hand mask failed on a real bottle recording; the presence score did not reliably distinguish a real grasp from a fake one.

SAM2 runs with two different prompts, producing two masks. Qwen performs the remaining held-object check. This keeps the pipeline simpler without claiming SAM3 is inferior for every task.

SAM2 overview · SAM3 overview

DINOv3 encoder consolidation

Decision made: keep DINOv2 and SigLIP. DINOv3 is not adopted.

Module 7 **Confirmed**

The aim is to simplify visual comparison and reduce model weight. DINOv3-S has 21.60M parameters and B 85.66M, versus 225.21M for the existing pair. The proposed replacement is not deployed.

Method	Same object across hands, at 0.82	Correct mesh decisions: historical 11 / all 15
Current DINOv2 + SigLIP	DINOv2: 3/4	SigLIP: 4/11 · 7/15
DINOv3-S	2/4	5/11 · 8/15
DINOv3-B	2/4	5/11 · 7/15
DINOv3-B foreground comparison	Not measured as a separate re-ID method	10/11 · 13/15; still two absent objects assigned incorrectly

The earlier test compared reused/self crops rather than independent hand-to-hand views, and some source sessions were empty or had wrong masks. That proxy could not establish real re-identification. The corrected evaluation restored the data and used independent crops; it is complete and supersedes that first test.

Foreground matching ranks known objects better, but still wrongly assigns an absent pen and remote. Next research needs independent held-out recordings, absent-object checks and reliable rejection of unknown objects; any adopted replacement must also preserve the identity pipeline's spatial/time checks. Keep DINOv2 and SigLIP, and leave mesh matching disabled meanwhile.

Corrected evaluation and methods

DINOv3 overview and code

Mesh selection and headset replay

Meshes can render in replay; the current task selection policy is a temporary shortcut.

Modules 7, 9 **Fixed for demo**

> **Update, 12 Oct:** No object from Batch 2 has a 3D model, so there is nothing new to serve. The library cannot grow until orbit models are made automatically. See the audit below.

- **Rendering:** demo sessions had meshes assigned manually. The clock bounding-box mismatch was corrected so Unity's shape check accepts the mesh.
- **Orbit ownership:** orbit replay uses its own registered reconstruction rather than another session's mesh.
- **Task shortcut:** every task object is currently served the latest mesh-bearing orbit registration, without checking object identity. Source orbit dimensions accompany the mesh; task tracks remain unchanged.
- **Matching disabled:** SigLIP retrieval achieved 4/11 correct historical decisions; correct and incorrect scores overlap. Automatic matching remains switched off.

- **Still needed:** validated identity and mesh-quality selection, mesh-backed pose estimation where appropriate, and headset checks of the latest API policy.

This patch joins display and storage; it does not make the borrowed mesh the right object or improve the recorded poses.

Grip-first orbit mesh generation

Works by hand on two objects. It does not start by itself, and the first test on a new orbit failed.

Modules 3, 4, 8 **Partially fixed**

> **Update, 12 Oct:** Nothing starts a reconstruction after an orbit upload, so none of the 11 orbits from Batch 2B has a 3D model yet. A first attempt on a new orbit (wooden clock) failed the fit check (0.42 against a bar of 0.70) because the mask covered only the dial. Flipping an object mid-orbit breaks the method. See the audit below.

Grip first, then slowly circle the object for the rest of the clip. Hands23 identifies the starting object; SAM2 follows its outline. Masked COLMAP matches visible features across the images and reconstructs a 3D point cloud. Open3D builds the surface, and Quest head poses set its scale in metres.

Candle and clock reconstructions are measured successes. This confirms the reconstruction method; it does not establish unattended upload-to-mesh automation for every session. The latest clock reconstruction was run and registered manually.

Route	Input and purpose	Project finding
Hunyuan3D-Omni	One reference image; generates plausible hidden surfaces	Poor remote-control result despite a clean photograph.
TRELLIS.2	One reference image; generates a textured mesh	Cleaner remote-control result; unseen surfaces and orientation still need checking.
Fuse3D	Several real views fused through a generative model	Strong front, thinner sides in the same comparison.
COLMAP + Open3D	Matches real image features across an orbit; reconstructs points, then a surface	Measured metric candle and clock meshes. Low-texture objects can still fail.

The earlier comparison is a small project trial, not a general model ranking. COLMAP is the measured orbit route, but the bottle and rolling pin failed at feature matching. The bowl had healthy pairwise matches and failed later in multi-view consistency.

Follow-ups: mesh selection and replay · orientation and interaction.

COLMAP tutorial and code

FoundationPose with a trusted mesh

The service runs and beat the old tracker on four clock clips. In the live worker it runs beside the old tracker and does not replace it yet.

Module 5b **Partially fixed**

> **Update, 12 Oct:** Our earlier description overstated how far FoundationPose was used in the live pipeline. It is implemented and now runs as a memory-capped service, but in the live worker it runs in shadow mode: the old point-cloud pose is still the one saved. On four clock clips FoundationPose beat the old tracker on every one. Both lose the clock when its underside faces the camera, and FoundationPose skips orbit recordings. See the audit below.

FoundationPose was already implemented. The recorded activation blockers were the wrong GPU being almost full from another job, and a mesh loader expecting OBJ while registered assets used GLB. Moving the service to a GPU with room and supplying the expected mesh enabled real registration and more than 100 successful tracking calls.

The bottle trial still had 9 large rotation jumps in 112 transitions, mostly flip-and-recover pairs. Hand occlusion was a likely cause, not a proven diagnosis. A working container does not prove perfect tracking.

Default live task processing remains model-free while mesh matching is disabled. Borrowing a mesh for rendering does not rerun FoundationPose.

Without a trusted mesh, the remaining fallback issue is PointCloudPCA.

FoundationPose overview and code

Episode naming endpoint

Names come from Qwen labels through a fixed term list. Anything outside the list shows as unidentified.

Modules 2, 6 **Partially fixed**

> **Update, 12 Oct:** Most Batch 2B objects (Rubik's cubes, book, controller, tape, cereal box) show as unidentified because their labels are not in the list. Qwen also mislabels some objects (the candle as "hand watch"). The plan is to name objects from Qwen's own description. See the audit below.

`{object_key}_{counter}_{YYMMDD}`

The server uses existing Qwen3-VL labels and, when available, canonical category or label data. It normalises supported terms, rather than asking another model to invent a task title. The counter

starts at 00 for each object-key/date pair and grows beyond 99. The capture date prefers device-local time.

Outcome	Example display name	Meaning
Waiting / processing	`unidentified_object_00_261005`	Pending; the object is not yet identified.
Finished, no object	`unidentified_object_00_261005`	Unidentified; distinct processing state even though the name has the same form.
One recognised object	`clock_00_261005`	A supported token from the existing label or canonical record.
Two different objects	`bottle.clock_00_261005`	Distinct tokens sorted alphabetically and joined with a dot.
Repeated object type	`clock_x2_00_261005`	Two tracked records with the same token.
Three different types	`bottle.candle.clock_00_261005`	Up to three distinct tokens remain explicit.
Four or more types	`multiobj4_00_261005`	Number counts distinct types, rather than total objects.
Unsupported label	`unidentified_object_00_261005`	A detected object can still have an unsupported name; check the state and label.
Device legacy title	`untitled`	Old device title, not a generated server naming outcome.

`GET /sessions/{uuid}/metadata` includes the display name, processing state, revision, timestamp and grasp result. Pending and finished-with-no-object remain distinct. Device folders are not renamed.

PointCloudPCA: pose fallback without a mesh

A partial view or noisy depth can make the estimated object position jump.

Module 5a **Partially fixed**

> **Update, 12 Oct:** Steadier-pose tests (W2, W3) cut the shake but locked in wrong directions and were not adopted. Direction should come from FoundationPose's mesh. See the audit below.

This route estimates the centre and orientation of the visible depth points. If the visible patch changes, the estimated object can suddenly move even when the real object does not. The full object shape is unknown.

> Slow candle: maximum frame-to-frame jump 0.153 m.

> Fast clock: maximum frame-to-frame jump 1.053 m.

> Table-contact Stanley cup: maximum jump 0.738 m.

These are recorded pre-guard measurements, not new results after the fix. During a confirmed grip, the deployed guard drops estimates more than 0.5 m from every available wrist/palm. Missing hand landmarks cannot be checked, and unheld objects are not covered by this rule.

Next: measure post-guard natural-motion performance and add justified motion, table-contact or temporal constraints. Robust fallback replay remains unconfirmed.

Automatic mesh matching

A first limited matcher is on for 3 objects. New objects have no model, so they cannot match.

Module 7 **Partially fixed**

> **Update, 12 Oct:** A first live matcher (reference-photo similarity, fixed settings) is now on. It only knows three objects with models: the ceramic clock, the scented candle and the rolling pin. In a replay of 107 objects, adding more reference recordings cut wrong matches from 8 to 0 (counted by reading, not confirmed labels); that larger library is not deployed. None of the 43 Batch 2B clips matched, because those objects have no model. See the audit below.

The full mesh-selection situation is collected in Mesh selection and headset replay. The limited matcher and the missing models are why most live tasks do not reach FoundationPose.

Replay orientation and interaction

Facing and grabbing are partly done on the Unity side. Several items are still open or unverified on the headset.

Module 9 **Partially fixed**

> **Update, 12 Oct (code review by the Unity side, not yet re-verified on the headset):**

- **Ghost replay faces the viewer at spawn:** still open. Replay is placed ahead of and below the head but its recorded direction is not turned to the viewer.

- **Clock orbit mesh orientation:** partly done. A wrong global rotation was removed and Graham confirmed the earlier clock replay on the headset. The slanted start of the later own-orbit clock mesh comes from the server-built mesh, so it cannot be fixed on the headset without a server value.
- **Placement and size of new recordings:** partly done. List placement, row scale and multi-window placement exist, but newly added recordings are unverified on the headset.
- **Direct grab and distance pointer:** partly done. Pinch-grab with wrist rotation and pointer buttons exist. There is no distance-pointer grab for moving a replay.
- **FoundationPose replay on the headset:** unverified. The earlier test showed only a wireframe, because the mesh was hidden by a shape check, so it does not confirm pose or mesh rendering.

Dedicated Maestro server endpoint

Awaiting IT approval. The request is submitted and there has been no response yet.

Module 2 **Not confirmed**

Maestro currently uses the shared university proxy. A dedicated endpoint is pending for EEAI 2026 and the presentation of the encapsulated edge-AI system. A dedicated endpoint is the intended infrastructure change, but successful operation after migration is not yet established.

After IT supplies it, validate headset URLs, HTTPS, uploads, metadata and mesh downloads together; changing the endpoint may expose integration problems.

Issues found since 5 Oct

Last week’s recordings surfaced new problems. They are being handled one at a time.

Modules 3, 4, 7, 8 **Not confirmed**

- **No 3D model is made automatically from an orbit.** 11 orbits from Batch 2B have none, and the first test on a new orbit failed.
- **Empty or split results:** 10 clips came out empty and several objects were split in two.
- **Flipping an object mid-orbit** breaks tracking and makes duplicates.
- **The clock underside** is lost by both trackers.
- **Names:** many objects show as unidentified.
- **Shared GPUs:** other jobs filled the GPU our live worker used. Maestro now has its own GPU and a lock so heavy jobs take turns.

Evidence and videos are in the audit below.

Possible feedback and design decisions

- **Naming convention:** should users keep the compact machine name, see a friendlier title, or add their own task title? Current names describe object records, not the action performed.
- **Offline Wi-Fi:** distinguish connected, unstable and fully offline states with an explicit icon/message. Agree how queued uploads and unavailable metadata should appear; client behaviour still needs validation.

~~Estimated Timeline~~

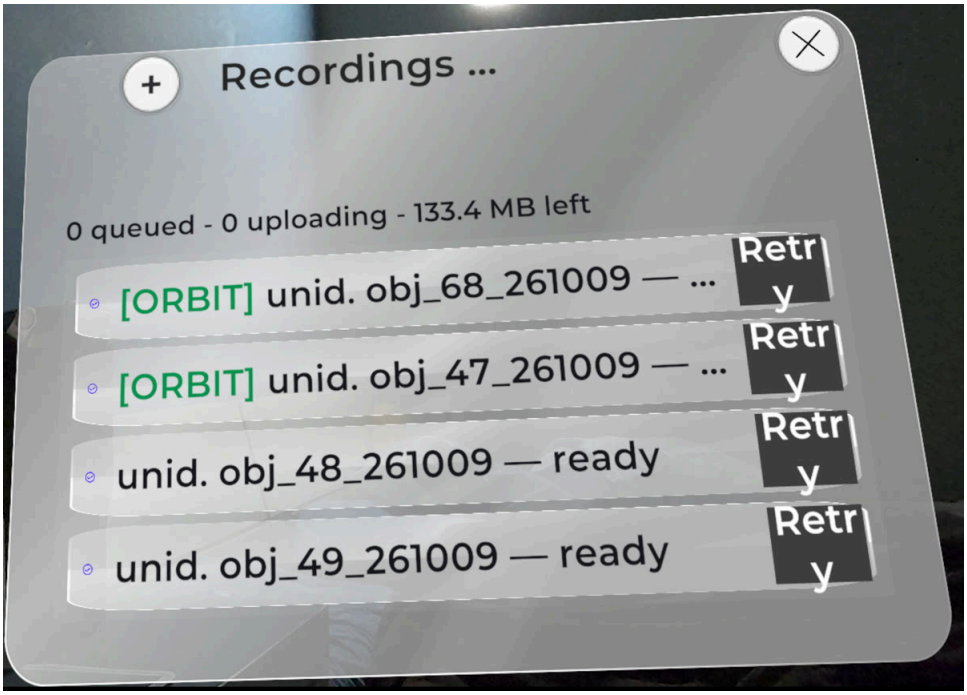
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Audit (12 October 2026)

Recordings

- Batch 2 was split into 2A and 2B. Uploads kept failing, so we built an upload queue first (first in, first out, with visible progress)
- 2B: 43 clips processed. None matched a 3D model, because no 3D models exist yet for the new objects. 10 came out empty and 4 split one object into two

Headset Recordings tab. "133.4 MB left" is one old failed clip being counted. A display fix is written and not yet installed



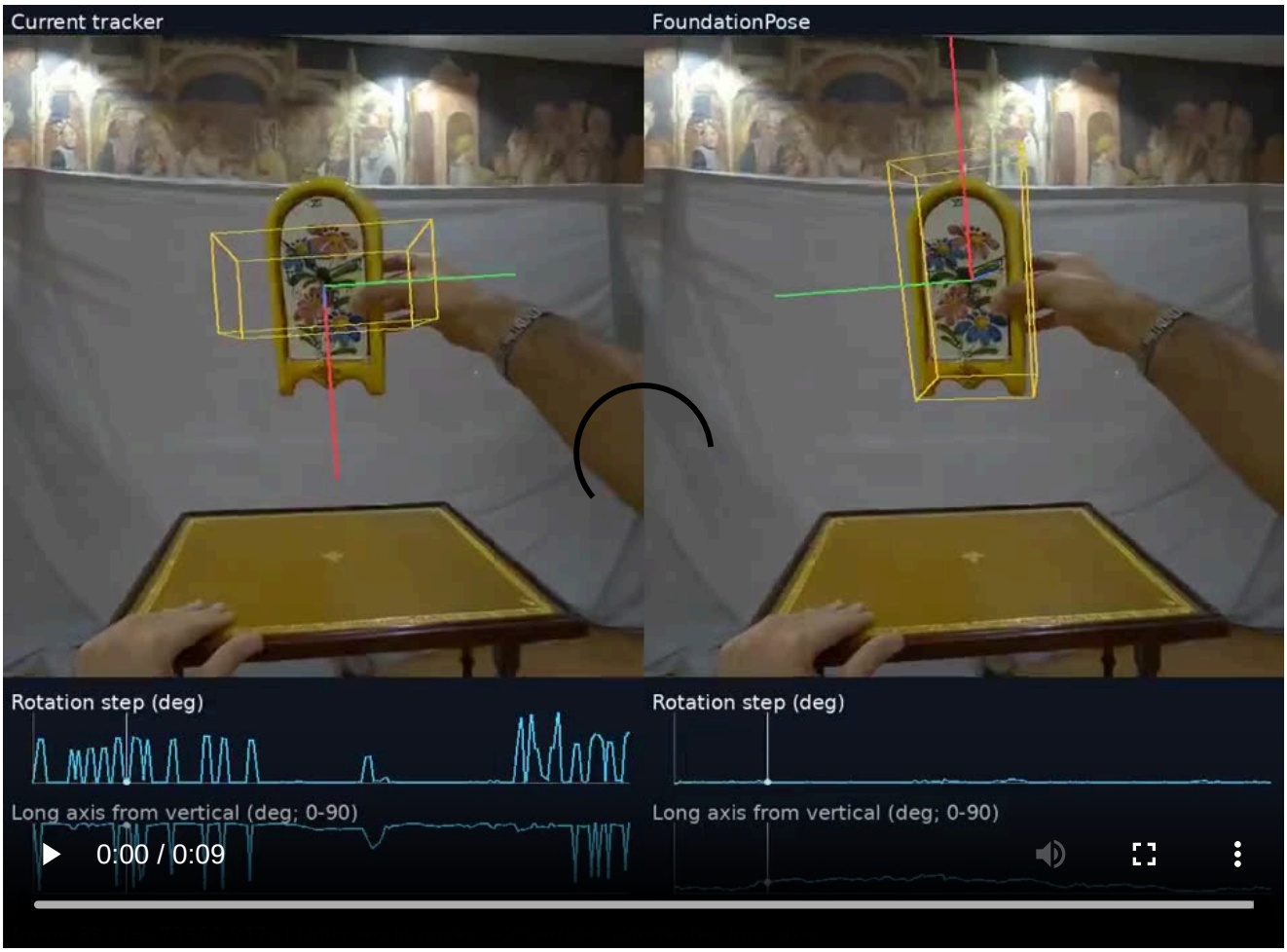
Old tracker vs FoundationPose

- FoundationPose tracks using the object's 3D model. The old tracker does not
- **No smoothing on either side**

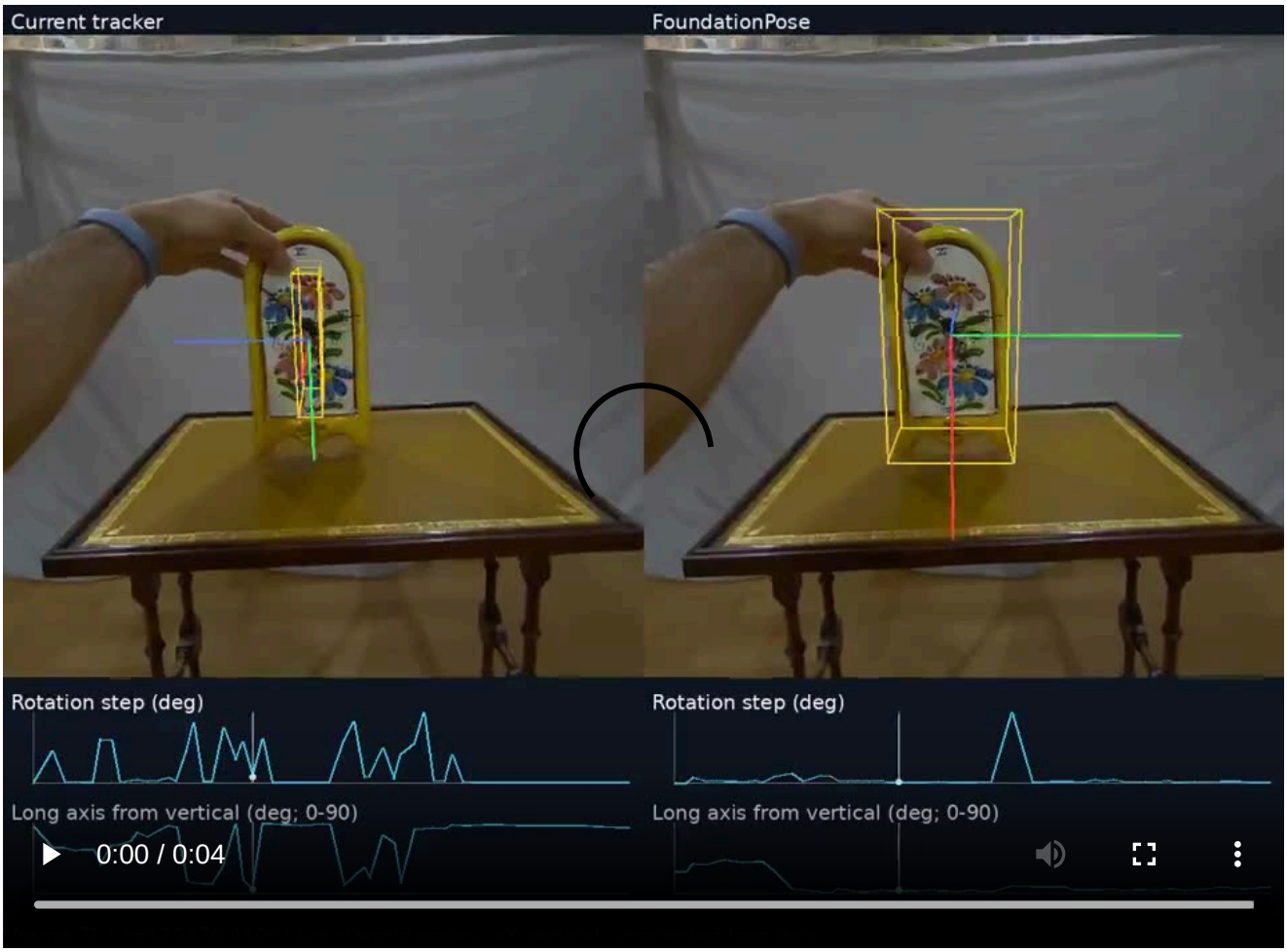
Clock, live clip. The old tracker spikes. FoundationPose stays upright



Clock, hand task



Clock, back view



Clock plus a second object. Shows the one-object-becomes-two problem



- **Problem: both trackers lose the clock when its underside faces the camera.** Likely an orbit issue, which we are investigating

Flipping the object mid-orbit

- The tracker loses the object and makes duplicates
- Flipping sometimes does not work at all: the candle and the stopwatch were not found
- The bottle kept the object but added 4 duplicates
- **Next: lock onto one object from the first grip. Merge the duplicates, and treat later changes as flips or rearrangements of that same object**
- **Aim: orbits that capture the whole object, underside included, so the 3D model and tracking cover it and the clock-underside failure above does not happen**

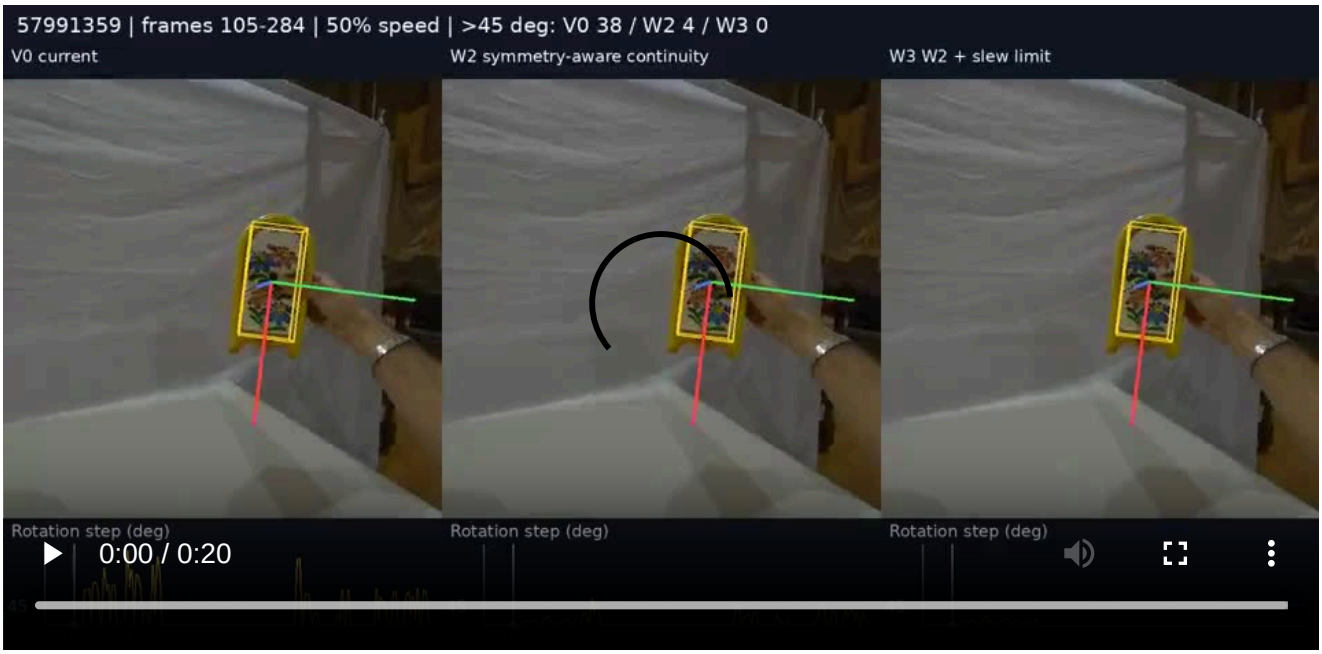
Bottle, flipped



Steadier pose (W2, W3)

- **W2** = symmetry-aware continuity (keeps the box from jumping between look-alike orientations)
- **W3** = W2 plus a speed limit on turning
- Done offline, alongside the FoundationPose work, on the old tracker's pose
- Result: much less shake, but it can lock in the wrong direction. On one clip it ended turned 90 degrees
- Not adopted for the old tracker
- **Open:** whether W2 or W3 help on top of FoundationPose. Not tested yet. FoundationPose's 3D model already fixes direction, so the gain may be small

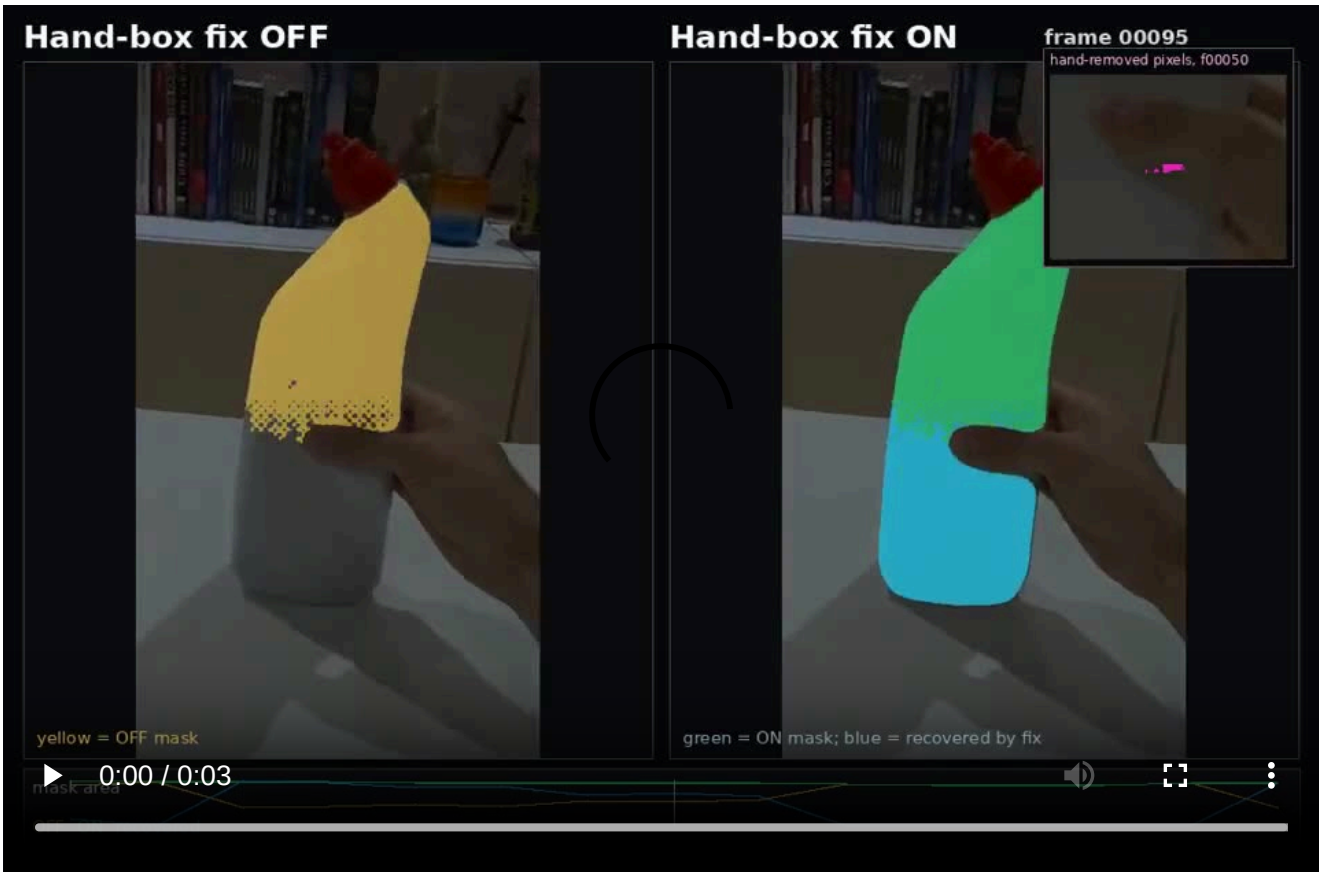
Current, W2, W3 on the same clip. Shake gone, W2 ends wrong



Cleaning the masks

- The mask is the outline of the object, with the hand cut out of it
- **Fix 1: cut the hand out using this frame's hand box.** Small gain. Fewer broken outlines. Not enough to adopt
- **Fix 2: restart the mask when it breaks.** No effect
- Still a problem: many clips never find the object or start badly. That needs a better start, not cleaning

Fix 1 off (left) and on (right), best case. The old version loses the lower half of the bottle (blue = what the fix recovers). The typical gain is much smaller



Matching

- Each clip is compared with reference recordings. The settings were not tuned
- More reference recordings helped. Wrong matches in a replay of 107 objects: 8, then 2, then 0 (counted by reading, not confirmed labels)
- Matching only helps once an object has a 3D model. That is why none of the 2B clips matched

Found since the audit

- Last week's recordings and iterations surfaced many new issues. They are being handled one at a time
- **No 3D models are made automatically from orbits.** The step was never built. 11 orbits are waiting
- **First test of making a model from a new orbit (wooden clock) failed.** Fit score 0.42, needed 0.70. The mask covered only the dial and only part of the orbit was usable. The method is being checked on an older good orbit
- **Many objects came out "unidentified".** Names come from a fixed list. They should come from Qwen's description
- The printed marker board (ChArUco) used in September is no longer needed for the 3D model route, which gets its scale from the headset's own movement

Next

- Orbit fix: lock onto one object from the first grip, merge duplicates, handle flips. This also fixes split objects and should cover the clock underside
- Make 3D models right after each orbit, other objects first. Each model is checked before it joins the library
- Name objects from Qwen

Still to investigate

- How to start tracking small objects that are never found (10 empty clips)
- Why the clock underside is lost: the model missing the underside, or the tracker
- Cleaning masks is still open

Technical Detail

Full technical audit and measurements

Graham Pellegrini | M.AI.ESTRO VR | System audit | Finalized 4 October 2026